Hand Detection

1. Course Content

- Run example programs, use the mediapipe framework to detect hand joint points

2. Program Startup

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

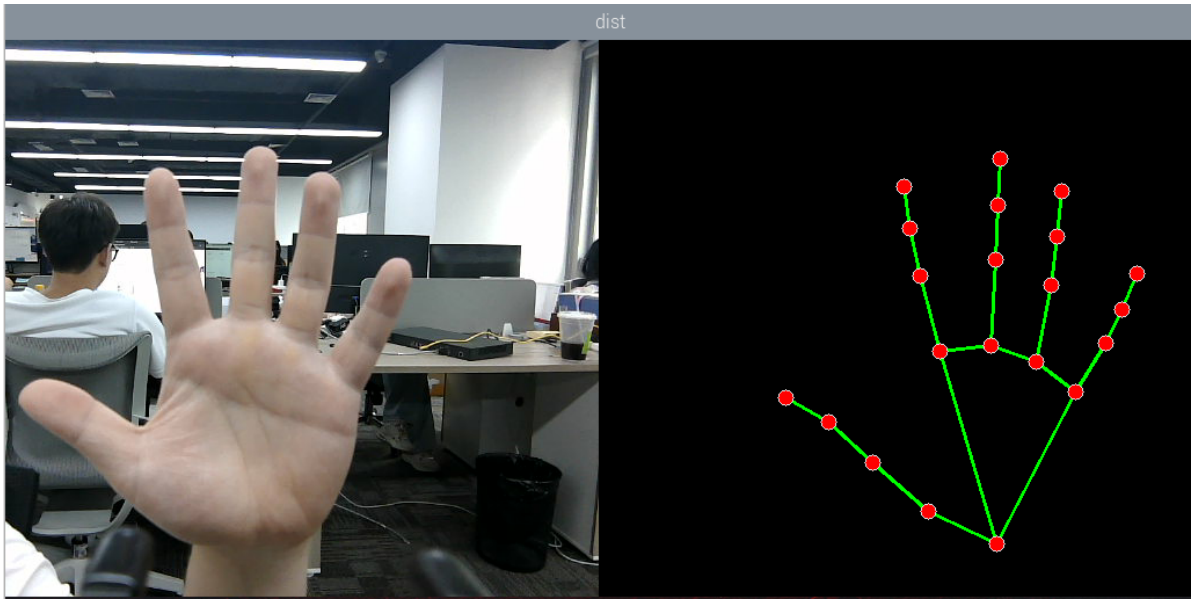
Open a terminal inside the container,

```
exec_m3
```

- Run hand detection command in terminal

```
ros2 launch m3_bringup mediapipe_demo.launch.py demo:=HandDetector
```

After the program runs, as shown in the figure below, the detected hand joint points will be displayed on the right side of the image



3. Source Code Analysis

- Program code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/mediapipe/01_HandDetect
or.py
```

```
import rclpy
from rclpy.node import Node
import mediapipe as mp
import cv2 as cv
import numpy as np
import time
import os
from cv_bridge import CvBridge
from sensor_msgs.msg import Image
from arm_msgs.msg import ArmJoints
import cv2
```

- Initialize data and define publishers and subscribers

```
    def __init__(self,name, mode=False, maxHands=2, detectorCon=0.5,
trackCon=0.5):
        super().__init__(name)
        #Parameter declaration and acquisition

 self.declare_parameter('rgb_topic','/ascamera_hp60c/camera_publisher/rgb0/image
')

 rgb_topic=self.get_parameter("rgb_topic").get_parameter_value().string_value
        self.qos_profile = QoSProfile(
            reliability=QoSReliabilityPolicy.BEST_EFFORT,
            history=QoSHistoryPolicy.KEEP_LAST,
            depth=1
        )
        self.pTime=0
        self.mpHand = mp.solutions.hands
        self.mpDraw = mp.solutions.drawing_utils
        self.hands = self.mpHand.Hands(
            static_image_mode=mode,
            max_num_hands=maxHands,
            min_detection_confidence=detectorCon,
            min_tracking_confidence=trackCon)
        self.lmDrawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 0,
255), thickness=-1, circle_radius=6)
        self.drawSpec = mp.solutions.drawing_utils.DrawingSpec(color=(0, 255,
0), thickness=2, circle_radius=2)
        #create a publisher
        self.rgb_bridge = CvBridge()
        self.sub_rgb =
self.create_subscription(Image,rgb_topic,self.get_RGBImageCallBack,self.qos_prof
ile)
```

- Color image callback function

```
    def get_RGBImageCallBack(self,msg):

        rgb_image = self.rgb_bridge.imgmsg_to_cv2(msg, "bgr8")
        frame, img = self.pubHandsPoint(rgb_image, draw=False)
        dist = self.frame_combine(frame, img)
        cTime = time.time()
        if self.pTime == 0:
            fps = 0
        else:
            fps = 1 / (cTime - self.pTime)
        self.pTime = cTime
        text = "FPS : " + str(int(fps))
        cv.putText(dist, text, (20, 30), cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0,
255), 1)
        key = cv2.waitKey(1)
        cv.imshow('dist', dist)
```

pubHandsPoint function:

```
    def pubHandsPoint(self, frame, draw=True):
        img = np.zeros(frame.shape, np.uint8)
        img_RGB = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
        self.results = self.hands.process(img_RGB)
        if self.results.multi_hand_landmarks:
            for i in range(len(self.results.multi_hand_landmarks)):
                if draw: self.mpDraw.draw_landmarks(frame,
self.results.multi_hand_landmarks[i], self.mpHand.HAND_CONNECTIONS,
self.lmDrawSpec, self.drawSpec)
                self.mpDraw.draw_landmarks(img,
self.results.multi_hand_landmarks[i], self.mpHand.HAND_CONNECTIONS,
self.lmDrawSpec, self.drawSpec)
```

frame_combine image merging function:

```
def frame_combine(slef,frame, src):
    #Check if the image is 3-channel, i.e., RGB image
    if len(frame.shape) == 3:
        #Get the size of the two images and stitch them together
        frameH, frameW = frame.shape[:2]
        srcH, srcW = src.shape[:2]
        dst = np.zeros((max(frameH, srcH), frameW + srcW, 3), np.uint8)
        dst[:, :frameW] = frame[:, :]
        dst[:, frameW:] = src[:, :]
    else:
        src = cv.cvtColor(src, cv.COLOR_BGR2GRAY)
        frameH, frameW = frame.shape[:2]
        imgH, imgW = src.shape[:2]
        dst = np.zeros((frameH, frameW + imgW), np.uint8)
        dst[:, :frameW] = frame[:, :]
        dst[:, frameW:] = src[:, :]
    return dst
```